

DATABASE MANAGEMENT SYSTEMS

II B. TECH- I SEMESTER								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A5CS05	PCC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES The course should enable the students to: 1. Discuss the basic database concepts, applications, data models, schemas and instances. 2. Design Entity Relationship model for a database. 3. Demonstrate the use of constraints and relational algebra operations. 4. Describe the basics of SQL and construct queries using SQL 5. Understand the importance of normalization in databases. 6. Demonstrate the basic concepts of transaction processing and concurrency control. 7. Understand the concepts of database storage structures and identify the access techniques.								
COURSE OUTCOMES At the end of the course the students are able to: 1. Use the basic concepts of Database Systems in Database design 2. Design a Database using ER Modelling 3. Apply normalization on database design to eliminate anomalies 4. Apply SQL queries and PL/SQL queries to interact with Database 5. Analyze database transactions and can control them by applying ACID properties.								
UNIT - I	INTRODUCTION						CLASSES: 09	
INTRODUCTION: Introduction and applications of DBMS, Purpose of data base, Data Independence, Database System architecture- Levels, Database users and DBA. DATABASE DESIGN: Database Design Process, ER Diagrams - Entities, Attributes, Relationships, Constraints, keys, extended ER features, Generalization, Specialization, Aggregation, Conceptual design with the E-R model.								
UNIT - II	RELATIONAL MODEL &SCHEMA REFINEMENT						CLASSES: 09	
THE RELATIONAL MODEL: Introduction to the relational model, Integrity constraints over relations, Enforcing integrity constraints, Querying relational data, Logical database design: E-R to relational, Introduction to views, Destroying/altering tables and views. SCHEMA REFINEMENT AND NORMAL FORMS: Introduction to schema refinement, functional dependencies, reasoning about FDs. Normal forms: 1NF, 2NF, 3NF, BCNF, properties of decompositions, normalization, schema refinement in database design.								
UNIT - III	RELATIONAL ALGEBRA AND CALCULS &SQL						CLASSES: 11	
RELATIONAL ALGEBRA AND CALCULUS:Relational algebra operators, relational calculus - Tuple and domain relational calculus. SQL: Basics of SQL, DDL,DML,DCL, structure – creation, alteration, defining constraints – Primary key, foreign key, unique, not null, check, IN operator, Functions - aggregate functions, Built-in functions – numeric, date, string functions, set operations, sub-queries, correlated sub-queries,								
UNIT - IV	SQL &PL/SQL						CLASSES: 11	

SQL: Use of group by, having, order by clauses, join and its types, Exist, Any, All clauses . Transaction control commands – Commit, Rollback, Save point, PL/SQL: Environment, block structure, variables, operators, data types, control structures; cursors, stored procedures, Triggers.		
UNIT - V	TRANSACTION & CONCURRENCY CONTROL	CLASSES: 10
TRANSACTIONS MANAGEMENT: Transaction concept, transaction state, concurrent executions, Serializability, recoverability, testing for serializability. CONCURRENCY CONTROL AND RECOVERY SYSTEM: Concurrency control, lock based protocols, time-stamp based protocols, validation based protocols, multiple granularity and deadlock handling. Recovery system - failure classification, storage structure, recovery and atomicity, log based recovery.		
TEXT BOOKS		
1. Abraham Silberschatz, Henry F. Korth, S. Sudharshan, "Database System Concepts", Sixth Edition, Tata McGraw Hill, 2011. 2. Raghurama Krishnan, Johannes Gehrke, "Data base Management Systems", TATA McGraw Hill, 3rd Edition, 2007. 3. R.P. Mahapatra & Govind Verma, Database Management Systems, Khanna Publishing House, 2013. 4. Michael McLaughlin, Oracle Database 11g PL/SQL Programming, Oracle press.		
REFERENCE BOOKS		
1. Peter Rob, Carlos Coronel, Database Systems Design Implementation and Management, 7th edition, 2009. 2. Scott Uman, Michael McLaughlin, Ron Hardman, "Oracle database 10g PL/SQL programming ", 6th edition, Tata McGraw Hill, 2010 3. S.K.Singh, "Database Systems Concepts, Design and Applications", First edition, Pearson Education, 2006. 4. Ramez Elmasri, Shamkant B. Navathe, "Fundamentals of Database Systems", Fourth Edition, Pearson / Addison wesley, 2007		
WEB REFERENCES		
1. http://www.learnadb.com/databases/how-to-convert-er-diagram-to-relational-database 2. https://www.w3schools.com/sql/sql_create_table.asp 3. http://www.edugrabs.com/conversion-of-er-model-to-relational-model/?upm_export=print 4. http://ssyu.im.ncnu.edu.tw/course/CSDB/chap14.pdf 5. http://web.cs.ucdavis.edu/~green/courses/ecs165a-w11/8-query.pdf		
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